

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

26th November 2019

Subject CP3 – Communication Practice

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Background information

You are an Actuary working in the customer service department of a life insurance company that specialises in pension business. Your products cater both to the accumulation stage and the annuity stage. One of the products offered by your Company, at the accumulation stage is a unit linked product which offers two funds. One of the funds invests in equity with the proportion ranging from 80% to 100% with the balance being invested in cash. The other fund invests in Government Securities and Corporate Bonds to the extent of 80% with the balance being in cash. Five years before maturity customers can choose the Life Stage option which moves all the future premiums and the amount in the equity fund to the debt fund. In case they do not choose this option the amounts continue to be in the funds chosen by the customer. The product offers flexibility to buy an annuity any time after age fifty five.

The product has a 2% allocation charge of every premium paid. In addition it has a fund management charge of 1.25% p.a. for the equity fund and 0.75% p.a. for the debt fund.

At the time of sale the intermediary carries out a suitability assessment of the product for the customer which takes into account the age, income and risk appetite. The customer chooses a target income and based on this information a target purchase price range is arrived. The premium that needs to be paid by the customer is then arrived at by considering the returns that the Company uses for illustrating the growth of the funds. In addition, the annuity amount that would be available with the target purchase price is also shown. This annuity amount is illustrated using annuity prices at the time of sale. The Benefit Illustration has sufficient information about the risks involved in meeting the target purchase price and the target annuity amount and includes necessary caveats that returns and annuity prices are not guaranteed.

The illustration shows the growth of the funds at different scenarios. However for the purpose of arriving at the target purchase price the scenario assumed is the one in which debt fund returns are illustrated at an appropriate low duration GSec yield and equity fund returns are illustrated with GSec yield plus 2%.

Annual statements are sent to the customer illustrating the current fund value and the fund value available at the end of the term. In addition the statements also contain the annuity rates which would be available currently based on the current age and fund value and the annuity at the end of the term based on the age at maturity and the projected fund value. The fund earning rates that are used for illustration are reviewed periodically and are set based on the prevailing yields. Customers can request for statements whenever they feel the need.

One of your customers had bought this plan at age 40 for a term of 25 years. The policy is now in the 21st year. Your company as is the practice sent an annual statement at the policy anniversary illustrating the fund value and the annuity that would be available. The customer has written to you as follows

Now that I am closer to retirement I have been going through the statements that are being sent to me quite carefully. I am also considering retiring early and therefore have looked at the annuity quotes with interest. About three months back I received the annual statement. At that time I was also asked to choose between the Life Stage Option and the Regular option. I however, did not get back on this. I recently asked for another statement and now I see that under both the options the fund that would be available to me at the end of the term has gone down significantly. I can also see that the fund that would be available to me under the LifeStage option is lower. I remember that the

options were explained to me but would like to understand them better before taking a decision. What is the advantage of the Life Stage option? I also see that annuity rates have gone down significantly. I would like to be assured that the statements and the annuity quotes are correct as it seems that I would have significantly less income than what I had anticipated. On the basis of your reply I may take a decision to buy an annuity immediately or wait for some time and continue to invest till the end of the term and then avail an annuity.

You look at the statements of the customer. You see that the fund returns have been much better than illustrated at the time of purchase because of which the fund at the end of the 20th year was higher than illustrated.

You also know that there is a recession looming in the economy due to which the interest rates have been dropping for some time and now have suddenly dropped by close to 50 bps in the last three months. Equity markets are expected to be volatile. Your company has taken the call to revise the rates that would be used for benefit illustrations. The current rates being used by the company are 5% for the debt fund and 7% for the equity fund. You also ascertain that returns in the last three months on debt fund is about 5% and equity fund is about 4%, lower than the assumptions of 5.5% and 7.5% used in the illustration as on 1 December 2019.

You also ascertain that annuity prices are revised every week due to interest rate movements. However, recently your company had revised prices also on account of revision of mortality assumptions. The company had been incurring losses over some years now as it was seeing lower annuitant mortality than allowed for in the pricing. This revision has also resulted in the annuity rates falling quite sharply. The most recent annuity quotes have the effect of both interest rates and revision of mortality assumptions.

The extracts of the various statements sent to the customer at different points in time are provided in the following sections. Premiums are payable at the beginning of the year. The actual fund values are at the date of statements. Projected fund values are at the end of the year.

Extracts from the statement sent to the customer at the time of the original policy.**Date: 1 December 1999 (Amounts in Rs.)**

Name of the customer: Ms. A Bhat					
Current age: 40 years , Age at vesting: 65 years					
Target purchase price range: Rs. 55 lakhs to Rs. 60 lakhs , Premium amount: Rs. 100,000 p.a.					
Life stage option- Equity fund and debt fund proportion for first 20 years : 75% : 25%					
Equity fund and debt fund proportion for last 5 years : 0% :100%					
Assumed return for equity fund: 8% p.a., Assumed return for debt fund: 6% p.a.					
Year	Premium	Premium allocation charge	Debt Fund Value	Equity Fund Value	Total fund at the end of the year
1	100,000	2,000	25,775	78,388	104,163
2	100,000	2,000	52,892	161,988	214,880
3	100,000	2,000	81,420	251,148	332,569
4	100,000	2,000	111,433	346,237	457,671
5	100,000	2,000	143,009	447,650	590,659
6 to 19					
20	100,000	2,000	870,983	3,093,360	3,964,343
21	100,000	2,000	4,273,788	-	4,273,788
22	100,000	2,000	4,599,340	-	4,599,340
23	100,000	2,000	4,941,836	-	4,941,836
24	100,000	2,000	5,302,160	-	5,302,160
25	100,000	2,000	5,681,238	-	5,681,238

Non Life Stage option- Equity fund and debt fund proportion throughout the term of the policy : 75% : 25%					
Assumed return for equity fund: 8% p.a., Assumed return for debt fund: 6% p.a.					
Year	Premium	Premium allocation charge	Debt Fund Value	Equity Fund Value	Total fund at the end of the year
1	100,000	2,000	25,775	78,388	104,163
2	100,000	2,000	52,892	161,988	214,880
3	100,000	2,000	81,420	251,148	332,569
4	100,000	2,000	111,433	346,237	457,671
5	100,000	2,000	143,009	447,650	590,659
6 to 19					
20	100,000	2,000	870,983	3,093,360	3,964,343
21	100,000	2,000	3,166,860	3,377,456	4,319,549
22	100,000	2,000	3,450,956	3,680,445	4,697,349
23	100,000	2,000	3,753,945	4,003,582	5,099,191
24	100,000	2,000	4,077,082	4,348,208	5,526,619
25	100,000	2,000	4,421,708	4,715,751	5,981,274

Extracts from statement sent at the end of the 20th year

Date: 1 December 2019 (Amounts in Rs.)

Under the Life Stage option all future premiums will be invested in the debt fund. The amounts in the equity fund will be transferred to the debt fund in two installments in the next two months.					
Assumed return for debt fund: 5.5% p.a.					
Year	Premium	Premium allocation charge	Debt Fund Value	Equity Fund Value	Total fund at the end of the year
			870,983	3,658,473	4,529,456
21	100,000	2,000	4,845,351		4,845,351
22	100,000	2,000	5,176,122		5,176,122
23	100,000	2,000	5,522,467		5,522,467
24	100,000	2,000	5,885,120		5,885,120
25	100,000	2,000	6,264,851		6,264,851

Under the Regular option all future premiums will continue to be invested in equity and debt funds as per the originally chosen proportion.					
Assumed return for debt fund: 5.5% p.a., Assumed return for equity fund 7.5% p.a.					
Year	Premium	Premium allocation charge	Debt Fund Value	Equity Fund Value	Total fund at the end of the year
			870,983	3,658,473	4,529,456
21	100,000	2,000	937,649	3,961,722	4,899,372
22	100,000	2,000	1,007,454	4,283,641	5,291,095
23	100,000	2,000	1,080,547	4,625,377	5,705,924
24	100,000	2,000	1,157,081	4,988,152	6,145,233
25	100,000	2,000	1,237,218	5,373,260	6,610,478

Extracts from statement sent three months later after the above statement.

Date: 1 March 2020 (Amounts in Rs.)

Under the Life Stage option all future premiums will be invested in the debt fund. The amounts in the equity fund will be transferred to the debt fund in two installments in the next two months.					
Assumed return for debt fund: 5% p.a.					
Year	Premium	Premium allocation charge	Debt Fund Value	Equity Fund Value	Total fund at the end of the year
			890,384	3,721,124	4,611,509
21			4,822,388		4,822,388
22	100,000	2,000	5,127,659		5,127,659
23	100,000	2,000	5,445,790		5,445,790
24	100,000	2,000	5,777,322		5,777,322
25	100,000	2,000	6,122,820		6,122,820

Under the Regular option all future premiums will continue to be invested in equity and debt funds as per the originally chosen proportion.					
Assumed return for debt fund: 5% p.a., Assumed return for equity fund 7% p.a.					
Year	Premium	Premium allocation charge	Debt Fund Value	Equity Fund Value	Total fund at the end of the year
			890,384	3,721,124	4,611,509
21			933,205	3,869,589	4,802,795
22	100,000	2,000	998,049	4,166,367	5,164,416
23	100,000	2,000	1,065,624	4,479,949	5,545,573
24	100,000	2,000	1,136,045	4,811,288	5,947,333
25	100,000	2,000	1,209,433	5,161,389	6,370,823

Summary of annuity rates quoted at different points of time based on actual fund values and projected values

Date	Age applicable for annuity quote	Purchase price (Rs) (1)	Annuity amount (Rs) (2)	Annuity rate % (2)/(1)%
1 December 1999	65	5,681,238	536,269	9.4%
	65	5,981,274	564,590	9.4%
1 December 2019	60	4,529,456	392,664	8.7%
	65	6,264,851	568,575	9.1%
	65	6,610,478	599,943	9.1%
1 March 2020	60	4,611,509	354,645	7.7%
	65	6,122,820	497,654	8.1%
	65	6,370,823	517,811	8.1%

- Q. 1)** Draft a reply to the customer in about 800 to 900 words. Your reply should address the customer's concerns about fund values and annuity rates. Your reply can include the advantages and disadvantages of the Life Stage option. However your reply should not have any advice. You may advise the customer to talk to her agent or financial planner before taking any decision. [90]
- Q. 2)** A technical person needs to make judgments about the relevance of information in communicating with a non-technical person. Comment about any such judgments that you exercised. In addition highlight any jargon that you avoided using in your reply. [5]
- Q. 3)** Comment about how you highlighted to the customer the uncertainties involved and the implications of any decisions that would be taken by the customer. [5]
